



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
14-02-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

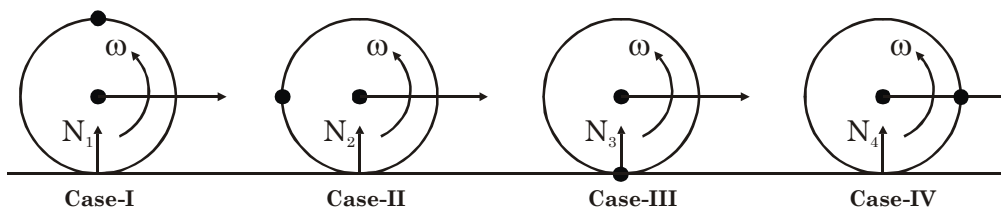
PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

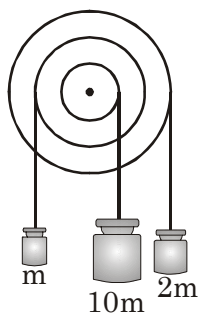
- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.
 - Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.
 - Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
 - Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).
 - Negative Marks* : -2 In all other cases.
 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
1. Two girls of equal mass m jump off from the border line of a stationary carriage of mass M with same horizontal velocity u relative to the carriage in same direction. Neglect the effect of friction. Mark **INCORRECT** options.
- (A) They will impart greater velocity to the carriage by jumping of simultaneously.
(B) They will impart greater velocity to the carriage by jumping one after the other.
(C) They will impart same velocity to the carriage in whatever manner they jump off.
(D) Insufficient data to reply.

Space for Rough Work

2. Ring of mass m and radius R is attached with a particle of mass m at its rim has 4 cases as shown. Centre of ring has some velocity, and angular velocity is $\omega = \sqrt{\frac{g}{R}}$. An external variable force couple acts on ring in such a way to keep angular velocity ω constant. When N_1 , N_2 , N_3 & N_4 are normal reactions between smooth ground and ring, then choose **CORRECT** alternatives.



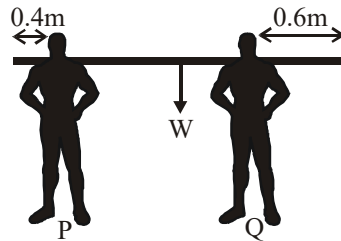
- (A) The value of N_1 is mg (B) The value of N_2 is $2mg$
 (C) The value of N_3 is $3mg$ (D) The value of N_4 is $4mg$
3. Three weightless pulleys with radii R , $2R$ and $3R$ are concentrically fastened together to make a single triple pulley system and mounted on an axis on which the triple pulley can rotate without friction (figure). The threads are light and they do not slip on the pulley. We suspended masses $10m$, m and $2m$ as shown.



- (A) Acceleration of $10m$ is $\frac{7g}{16}$ downwards (B) Acceleration of $2m$ is $\frac{21g}{16}$ downwards
 (C) Acceleration of m is $\frac{3g}{8}$ upwards. (D) Angular acceleration of pulley is infinite.

Space for Rough Work

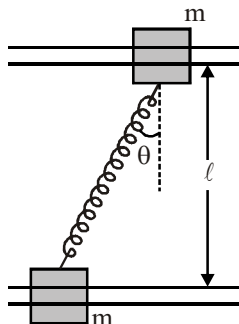
4. Two boys P and Q support a uniform plank of length 2m and weight (W) on their shoulders. If P's shoulder and Q's shoulder are 0.4 m and 0.6 m apart from the two ends respectively, what can be the supporting forces by the two boys Q & P respectively ?



- (A) 6 N, 4 N (B) 30 N, 20 N (C) 45 N, 30 N (D) 30 N, 60 N
5. Two sphere A and B of equal mass are free to move on smooth horizontal surface. A and B move towards each other with velocity vectors $a\hat{i} + b\hat{j}$ and $c\hat{i} + d\hat{j}$ respectively and collide when the line joining their centres is parallel to $\hat{i}$. After impact A and B have velocity vectors $p\hat{i} + q\hat{j}$ and $r\hat{i} + s\hat{j}$ respectively. The coefficient of restitution between the spheres is e (< 1)
- (A) $b = q$ (B) $c = r$ (C) $a + c = p + r$ (D) $ea = p$

Space for Rough Work

6. The figure below is the TOP-VIEW of two equal masses 'm' that can slide over frictionless rails on a horizontal floor. The rails are separated by distance ℓ as shown. The masses are connected with a massless spring of natural length ℓ and spring constant K. At the initial moment of time the masses are at rest.



- (A) The maximum magnitude of relative velocity between the two masses is given by

$$\ell \left(\frac{1 - \cos \theta}{\cos \theta} \right) \sqrt{\frac{2K}{m}}$$

- (B) The maximum magnitude of relative velocity between the two masses is given by

$$\ell \left(\frac{1 - \cos \theta}{\sin \theta} \right) \sqrt{\frac{K}{2m}}$$

- (C) As the spring comes to its natural length it is cut suddenly from the center. The speed of

lower block as shown in figure just after the spring is cut will be $\frac{\ell}{4} \sqrt{\frac{K}{2m}}$ (Take $\theta = 37^\circ$)

- (D) As the spring comes to its natural length it is cut suddenly from the center. The speed of

lower block as shown in figure just after the spring is cut will be $\frac{\ell}{3} \sqrt{\frac{2K}{m}}$ (Take $\theta = 37^\circ$)

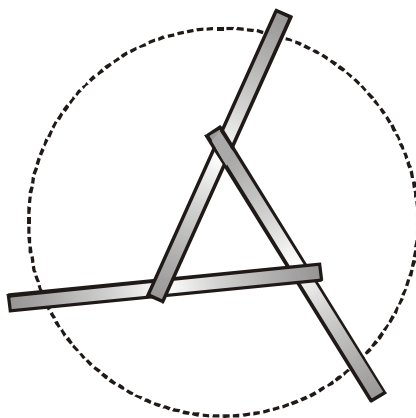
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

On the edge of a large vase (indicated by dashed line in figure which is top view) standing on horizontal table, rest three identical long and thin sticks each which has mass m distributed uniformly along the entire length. The ends of each stick touch the vase and the midpoint of another stick.

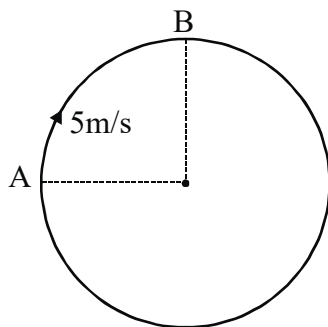


7. What is the force exerted by each rod on the vase ?
 (A) mg (B) $\frac{4mg}{3}$ (C) $\frac{3mg}{2}$ (D) $\frac{mg}{3}$
8. What is the force exerted by a rod on another rod?
 (A) mg (B) $\frac{3mg}{2}$ (C) $\frac{mg}{3}$ (D) $\frac{4}{3}mg$

Space for Rough Work

Paragraph for Questions 9 and 10

Two friends A and B (each weighing 40 kg) are sitting in a frictionless well having vertical circular smooth wall. They subtend angle 90° at the centre of well. 'A' rolls a ball of mass 40kg in clockwise direction on the platform base towards B along the wall of the well which B catches. Then similarly B rolls the ball towards 'A' anticlockwise and A catches it. The ball has a fixed speed of 5 m/s. (Neglect catching and throwing time).



9. Find the total impulse by the wall of well on whole system by the time ball travels from A to B
(A) 200 kg-m/s (B) 400 kg-m/s (C) zero (D) None of these
10. Find the angular displacement of A till he meets B
(A) 150° (B) 60° (C) 90° (D) 180°

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

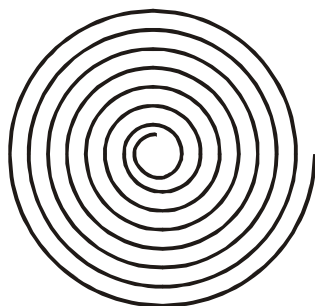
-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

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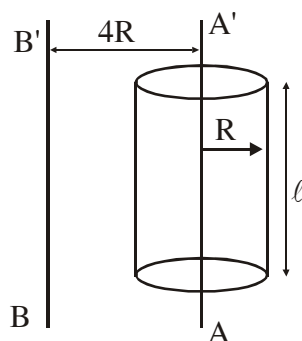
- Three particles A, B, C of masses m , $2m$, $3m$ respectively lie at rest in that order in a straight line on a smooth horizontal table. The distance between consecutive particles is a . A slack light inelastic string of length $2a$ connects A and B. An exactly similar slack string of length $2a$ connects B and C. If A is projected in the direction CBA with speed v . The total loss of kinetic energy when C has started to move is $(n) mv^2$. Find the value of n .

Space for Rough Work

2. Under action of a certain central force field, a particle of mass m undergoes motion along the curve $r = a\theta$. Here a is a positive constant. The magnitude of the angular momentum of the particle about the centre of the field is L . Calculate the potential energy (in J) at $r = 1\text{m}$ under which such a motion is possible. Take total energy = 100 J, $L = 5 \text{ kg m}^2/\text{s}$, $m = 1\text{kg}$, $a = 0.4\text{m}$, find potential energy at $r = 1\text{m}$.

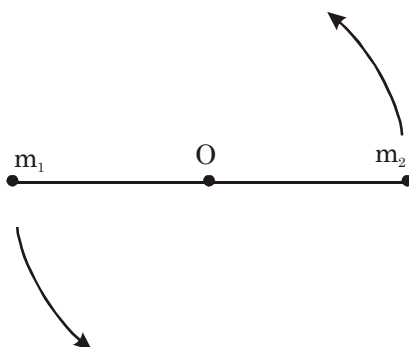


3. Consider a thin cylinder of mass m , radius R and length ℓ . AA' is axis through centre and BB' is at a distance $4R$ parallel to AA' . Now cylinder is heated uniformly for a long time. Its radius becomes $\frac{3}{2}R$. Change in moment of inertia along BB' due to heating is nmR^2 . Find the value of n .

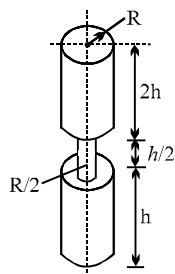


Space for Rough Work

4. A light rod of length $\ell = 1\text{m}$ with two balls on the ends (their masses are m_1 and m_2 ; $m_1 = 100\text{gm}$; $m_2 = 50\text{ gm}$) can rotate around a horizontal axis passing through the middle of the rod (see picture). Rod is brought to a horizontal position and released. With what force F (in Newton) does the rod press on the axis just after the release?

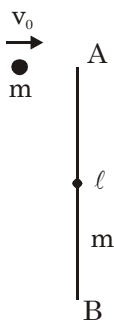


5. A merry-go-round is a common piece of playground equipment. A 4 m diameter merry-go-round with a moment of inertia of $500\text{ kg}\cdot\text{m}^2$ is spinning at 5.6 rad/s . John runs tangent to the merry-go-round at 5 m/s , in the same direction that it is turning, and jumps onto the outer edge. John's mass is 30 kg . What is the merry-go-round's angular velocity, (in rad/s), after John jumps on?
6. Two cylinders one of height h and the other of height $2h$, each having a radius R are joined by a smaller concentric cylinder of half the radius and height $h/2$ made of the same material. The centre of mass of system is $(x)h$ above the centre of the smallest cylinder. Then the value of x is.



Space for Rough Work

7. A mass m travels at speed v_0 perpendicularly to a stick of mass m and length ℓ , which is initially at rest. The mass collides completely inelastically with the stick at one of its end, and sticks to it. The location of point from end A (in cm) which is at instantaneous rest just after collision is X. (take : $\ell = 1.2$ meter)



8. A system of particles consists of several particles. Total mass of all the particles is 4 kg. To apply Newton's laws of motion in center of mass frame to one of the particles of mass 1 kg, you have to assume a pseudo force of $(0.3\hat{i} - 0.4\hat{j})$ N acting on it. What is the magnitude of the net external force (SI units) acting on the whole system?

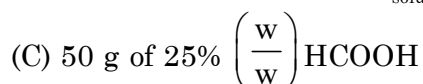
Space for Rough Work

PART-2 : CHEMISTRY

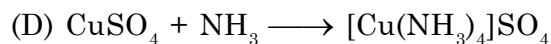
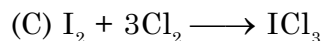
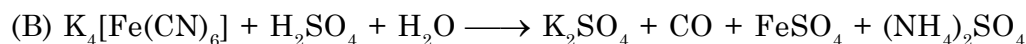
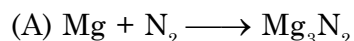
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1. Solutions containing 23g HCOOH is/are :



2. Which of the following is/are not redox reaction(s) ?



Space for Rough Work

3. Two identical vessels are connected by a narrow tube of negligible volume. Initially, the vessels were containing a total of 0.7 mole of an ideal gas at 27°C and 1.05 bar. Now, one vessel is heated to 227°C and other is cooled to -73°C . Which of the following is/are correct information(s) ? ($R = 0.083$ litre-bar/K-mol)
- (A) Volume of each vessel is 8.3 litre.
 (B) Final pressure of gas in both vessel is 1.0 bar.
 (C) Final mole of gas in the vessel at 227°C is 0.5.
 (D) Initial mole of gas in each vessel was 0.35.
4. Which of the following process is/are endothermic.
- (A) $\text{N} \rightarrow \text{N}^{-}$ (B) $\text{P} \rightarrow \text{P}^{-}$ (C) $\text{O} \rightarrow \text{O}^{-}$ (D) $\text{S} \rightarrow \text{S}^{2-}$
5. Which of the following is/are **CORRECT** for first Pnicogen element.
- (A) exist as diatomic molecule (B) gas at room temperature
 (C) colourless (D) odourless
6. Which of the following statement(s) is/are **CORRECT** ?
- (A) In a p-orbital the number of nodal planes equal to the azimuthal quantum number
 (B) The electronic energy of the H-atom depends only on n and is independent of ℓ and m
 (C) Those atoms in which all electrons are paired slightly repelled by an external magnetic field and are said to be paramagnetic
 (D) Those atoms in which one or more electrons have unpaired spins are strongly attracted to magnetic field and are said to be diamagnetic

Space for Rough Work

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Paragraph for Questions 7 and 8

Consider the following oxoacids of phosphorus : H_3PO_4 , H_3PO_3 and H_3PO_2 . All 'P' atoms in the oxoacids and oxoanions are 4 coordinated and contain at least one P=O unit. All 'P' atoms in oxoacids have at least one P-OH group which ionises to give H^+ ion. The oxidation state of 'P' is +5 only when it is directly attached to 4 oxygen atoms. For every two units decrease in oxidation state of Phosphorus, there is one P-H bond.

7. The number of H-atoms directly attached to 'P' in one molecule of each of H_3PO_4 , H_3PO_3 and H_3PO_2 are respectively :
(A) 0, 1 and 2 (B) 1, 0 and 2 (C) 2, 1 and 0 (D) 0, 2 and 1
8. The number of acidic hydrogens present in one formula unit of each of NaH_2PO_4 , NaH_2PO_3 and NaH_2PO_2 are respectively :
(A) 0, 1 and 2 (B) 1, 0 and 2 (C) 2, 1 and 0 (D) 2, 0 and 1

Space for Rough Work

Paragraph for Questions 9 and 10

A solid vessel contain 7.1 g chlorine gas at ambient pressure at T K. The vessel is kept in a bath which is 30° hotter. In order to maintain the original pressure, some gas is withdrawn and is found to occupy 246 mL at 1 atm at 27°C . [$R = 0.082 \text{ atm-litre/mol-K}$]

9. If the gas is not withdrawn the pressure increases by 0.11 atm, then what would be the original (ambient) pressure.
- (A) 1.22 atm (B) 0.66 atm
(C) 0.75 atm (D) 0.99 atm
10. Find out the volume of the vessel.
- (A) 2.24 litre (B) 3.12 litre
(C) 2.56 litre (D) 1.78 litre

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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- The answer to each question is a **NUMERICAL VALUE**.
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+		-	
0	0	0	0
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2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A container of capacity 20 L contains $N_2(g)$, water vapour and negligible but sufficient amount of liquid water to maintain equilibrium, at 300K and 1 atm. The liquid water is now decomposed completely into $H_2(g)$ and $O_2(g)$, by any means, without changing the temperature. If the final pressure becomes 1.86 atm, the mass of liquid water present initially (in gm) was (Aqueous tension at 300K = 0.04 atm, $R = 0.08 \text{ L-atm/K-mole}$)
2. An organic compound $C_xH_{2y}O_y$ (molar mass = 150 gm/mol) was burnt with twice the amount of oxygen needed for complete combustion to CO_2 and H_2O . The hot gases when cooled to 0°C and 1 atm pressure measured 2.24 litres. The water collected during cooling weighed 0.9 g.

Find the value of $\left(\frac{x+2y}{2}\right)$

Space for Rough Work

3. A sample of $\text{Fe}_2(\text{SO}_4)_3$ and FeC_2O_4 was dissolved in dil. H_2SO_4 . The complete oxidation of reaction mixture required 40 mL of N/16 KMnO_4 . After the oxidation, the reaction mixture was reduced by Zn and dil. H_2SO_4 . On again oxidation by same KMnO_4 , 60 mL were required. If the ratio of moles of $\text{Fe}_2(\text{SO}_4)_3$ and FeC_2O_4 in mixture is $x : y$ then find the value of x/y
- 4 In a spherical glass flask A of radius 1.0 m , containing 300 g of H_2 (g) , there was a rubber balloon (B) containing some N_2 (g). Inside B , there was another rubber balloon C containing some oxygen gas. At 27 °C , it was found that the balloon B had radius 60 cm and that of C was 30 cm . Calculate the total weight of gas (in gm) inside the flask .
5. Find the number of π bonds in the lewis dot structure of SO_4^{2-} .

Space for Rough Work

6. Find the number of chemical species having greater boiling point than He
 H_2 , Ne, Ar, Kr, SiO_2 , NaCl, T_2 , D_2

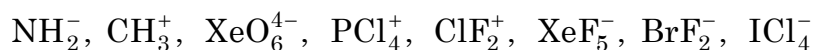
7. Calculate $\frac{(x+y)}{2}$ for tetrathionate ion.

where;

x = Maximum number of 'S' atoms must lie in same plane.

y = Maximum number of 'O' atom which shows same oxidation state.

8. Calculate $\frac{(x+y)}{z}$ for given molecular species.



where; x = Number of molecular species in which central atom carries two lone pair.

y = Number of hypovalent species

z = Number of super octet species

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $D = \begin{vmatrix} 1 & (\alpha + \beta) & (\alpha + \beta)^2 \\ 1 & (\beta + \gamma) & (\beta + \gamma)^2 \\ 1 & (\gamma + \alpha) & (\gamma + \alpha)^2 \end{vmatrix}$, where α, β, γ are positive integers such that $\alpha > \beta > \gamma$, then -

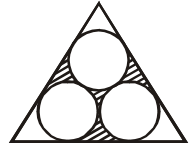
- (A) minimum value of D is 2 (B) minimum value of D is 1
 (C) D is always an odd integer (D) D is always an even integer

2. In triangle ABC, $\cos A + 2\cos B + \cos C = 2$, then -

- (A) $\tan \frac{A}{2} \tan \frac{C}{2} = 3$ (B) $\cot \frac{A}{2} \cot \frac{C}{2} = 3$ (C) $\cot \frac{A}{2} + \cot \frac{C}{2} = 2\cot \frac{B}{2}$ (D) $\tan \frac{A}{2} \tan \frac{C}{2} = 0$

Space for Rough Work

3. If I be the incentre and G be the centroid of triangle ABC, such that $\sin A : \sin B : \sin C$ is 4 : 7 : 9, then
 (A) $a : b : c$ is equal to 4 : 7 : 9
 (B) area of $\triangle BIC$: area of $\triangle CIA$: area of $\triangle AIB$ is equal to 4 : 7 : 9
 (C) area of $\triangle BGC$: area of $\triangle CGA$: area of $\triangle AGB$ is equal to 1 : 2 : 3
 (D) if L_1, L_2, L_3 are the lengths of the tangents drawn from vertices A,B,C respectively on incircle then $L_1 : L_2 : L_3$ is equal to 6 : 3 : 1
4. Three mutually tangent circles of equal radius 2 are as shown in the figure.
 The area of shaded portion is -
 (A) $4\sqrt{3} - 2\pi$ (B) $4(4\sqrt{3} - 2\pi)$
 (C) $4(6 + \sqrt{3} - 2\pi)$ (D) none of these
5. Consider the system of equations $(\lambda + 1)x + y + \lambda z = 3$, $\lambda x + 2\lambda y + z = 4$, $3x + (\lambda - 1)y + \lambda z = 2$, $\lambda \in \mathbb{R}$ then identify the correct statement/s-
 (A) there are exactly three values of λ for which system has unique solution
 (B) there are infinite values of λ for which system has unique solution
 (C) there is no value of λ for which system has unique solution
 (D) there is no value of λ for which system has exactly two solutions
6. If the equations $(a + 2)x^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root where $a, b, c \in \mathbb{N}$ then-
 (A) $b^2 - 4ac < 0$ (B) minimum value of $a + b + c$ is 16
 (C) $b^2 < 4ac + 8c$ (D) minimum value of $a + b + c = 7$



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

All the entries of determinant D_1 of order three are prime number, such that

$$a_{11} < a_{12} = a_{21} < a_{31} = a_{22} = a_{13} < a_{32} = a_{23} < a_{33} \quad \text{.....(i)}$$

one of the entries of this determinant is an even prime number and $a_{33} = 11$.

D_2 is another determinant whose least entry is 1, it satisfies (i), none of its entries are prime and all are natural numbers.

On the basis of above information, answer the following :

7. If S_1 denotes the sum of elements of D_1 and Δ_1 the value of D_1 , then -
(A) $S_1 + \Delta_1 = 50$ (B) $S_1 + \Delta_1 = 46$ (C) $S_1 + \Delta_1 = 34$ (D) $S_1 + \Delta_1 = 86$
8. If S_2 denotes the sum of elements of D_2 , then minimum value of S_2 is -
(A) 52 (B) 14 (C) 3 (D) 48

Space for Rough Work

Paragraph for Questions 9 and 10

If $a_i > 0$, $i = 1, 2, 3, \dots, n$ and $m_1, m_2, m_3, \dots, m_n$ be positive rational numbers, then

$$\left(\frac{m_1 a_1 + m_2 a_2 + \dots + m_n a_n}{m_1 + m_2 + \dots + m_n} \right) \geq \left(a_1^{m_1} a_2^{m_2} \dots a_n^{m_n} \right)^{1/(m_1 + m_2 + \dots + m_n)} \geq \frac{(m_1 + m_2 + \dots + m_n)}{\frac{m_1}{a_1} + \frac{m_2}{a_2} + \dots + \frac{m_n}{a_n}}$$

is called weighted mean theorem

where $A^* = \frac{m_1 a_1 + m_2 a_2 + \dots + m_n a_n}{m_1 + m_2 + \dots + m_n}$ = Weighted arithmetic mean

$G^* = \left(a_1^{m_1} a_2^{m_2} \dots a_n^{m_n} \right)^{1/(m_1 + m_2 + \dots + m_n)}$ = Weighted geometric mean

and $H^* = \frac{m_1 + m_2 + \dots + m_n}{\frac{m_1}{a_1} + \frac{m_2}{a_2} + \dots + \frac{m_n}{a_n}}$ = Weighted harmonic mean

i.e., $A^* \geq G^* \geq H^*$

Now, let $x^2 y^3 = 243$ ($x > 0, y > 0$)

On the basis of above information, answer the following questions :

9. The least value of $x^2 + 3y + 1$ is -
 (A) 15 (B) greater than 15 (C) 3 (D) less than 15
10. Which statement is correct -

- (A) $\frac{2x+3y}{5} \geq 3 \geq \frac{5}{\frac{3}{x} + \frac{2}{y}}$ (B) $\frac{2x+3y}{5} \geq 3 \geq \frac{5xy}{3x+2y}$
- (C) $\frac{2x+3y}{5} \geq 3 \geq \frac{5xy}{3x+4y}$ (D) $\frac{2x+3y}{5} \geq 3 \geq \frac{5xy}{2x+3y}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

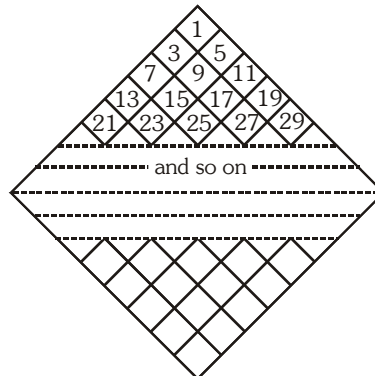
+										-									
●					●					●					●				
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $a679b$ is a five digit number that is divisible by 72, then $(a + b)$ is equal to
2. The medians of a $\triangle ABC$ are 9 cm, 12 cm and 15 cm respectively. Then the area of the $\triangle ABC$ is (in sq. cm.)-

Space for Rough Work

3. If a square of 12×12 units² is divided in 144 squares of 1×1 unit² and then terms of an A.P. are 1, 3, 5, 7,..... are filled up as shown in the figure, then sum of elements of both diagonals will be -



4. In a class of 58 students, 38 student got selected in PET exam, 20 selected in AIEEE and 15 selected in IITJEE.If only three students got selected in all three exams, then how many students got selected in exactly two of the three exams.
(each student is selected in atleast one exam out of these three)
5. Let $x_1 = 97$, $x_2 = \frac{2}{x_1}$, $x_3 = \frac{3}{x_2}$, $x_4 = \frac{4}{x_3}$,....., $x_8 = \frac{8}{x_7}$ then $\log_{3\sqrt{2}}\left(\prod_{i=1}^8 x_i - 60\right) =$

Space for Rough Work

6. Given that α, γ are roots of equation $Ax^2 - 4x + 1 = 0$ and β, δ the roots of the equation $Bx^2 - 6x + 1 = 0$. Find value of $(B - A)$ such that $\alpha, \beta, \gamma, \delta$ are in H.P.
7. Possible number of ordered pairs (x, y) satisfying the in equation $2^{(\sin x + \cos x)\sqrt{1+2y-y^2}} \geq 2^{y^2 + \frac{1}{y^2}}$ where $x \in [-3\pi, 3\pi]$
8. Solution set of simultaneous inequations $|x^2 - 4x|^{x^2 - 2x} > 1$ & $|x^2 - 2x|^{x^2 - 4x} > 1$ & $x < 4$ is given by $(-\infty, 1 - \sqrt{a}) \cup (1 + \sqrt{b}, 2 + \sqrt{c})$, then $a + b + c$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator